Lab#14 Create a quiz application that displays one question at a time. Track the user's score and show the result at the end.

Source Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Quiz App</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet"/>
  <style>      body { background: #f4f6fb; min-height: 100vh; display: flex; align-items: center;
justifycontent: center; font-family: 'Segoe UI', sans-serif; }
    .quiz-card { max-width: 620px; width: 100%; background: #fff; border-radius: 16px; border: 1px
solid #e2e8f0; padding: 2.5rem 2rem; }
    .progress { height: 8px; border-radius: 99px; margin-bottom: 1.75rem; }
    .progress-bar { background: #4f46e5; border-radius: 99px; transition: width 0.4s ease; }     .q-meta {
display: flex; justify-content: space-between; align-items: center; margin-bottom: 1rem; }
    .q-label { font-size: 13px; color: #64748b; }
    .score-pill { font-size: 13px; font-weight: 600; color: #166534; background: #dcfce7; padding: 3px
12px; border-radius: 99px; }
    .q-text { font-size: 1.15rem; font-weight: 600; color: #1e293b; margin-bottom: 1.5rem; lineheight:
1.5; }
    .opt-btn { width: 100%; text-align: left; background: #fff; border: 1.5px solid #e2e8f0; borderradius:
10px; padding: 0.75rem 1rem; font-size: 0.95rem; color: #334155; cursor:
pointer; display: flex; align-items: center; gap: 12px; transition: border-color 0.15s, background 0.15s;
margin-bottom: 10px; }
    .opt-btn:hover:not(:disabled) { background: #f8fafc; border-color: #94a3b8; }
    .opt-btn.correct { background: #f0fdf4; border-color: #16a34a; color: #166534; }
    .opt-btn.wrong  { background: #fef2f2; border-color: #dc2626; color: #991b1b; }
    .opt-btn:disabled { cursor: default; }
    .key-badge { display: inline-flex; align-items: center; justify-content: center; width: 28px; height:
28px; border-radius: 7px; background: #f1f5f9; border: 1px solid #e2e8f0; font-size:
12px; font-weight: 600; color: #64748b; flex-shrink: 0; }
    .opt-btn.correct .key-badge { background: #dcfce7; border-color: #16a34a; color: #166534; }
.opt-btn.wrong .key-badge  { background: #fee2e2; border-color: #dc2626; color: #991b1b;
}
    .feedback-box { min-height: 36px; font-size: 14px; margin: 0.75rem 0; display: flex; alignitems:
center; gap: 8px; }
```

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  .feedback-box.correct { color: #166534; }
  .feedback-box.wrong  { color: #991b1b; }
  .next-btn { display: none; }
  .result-emoji { font-size: 52px; display: block; margin-bottom: 0.5rem; }
  .stat-box { background: #f8fafc; border-radius: 10px; padding: 1rem; text-align: center; }
 .stat-val { font-size: 1.6rem; font-weight: 700; color: #1e293b; }
 .stat-lbl { font-size: 12px; color: #64748b; margin-top: 2px; }
  .review-item { padding: 0.6rem 0; border-bottom: 1px solid #f1f5f9; }
  .review-item:last-child { border-bottom: none; }
  .dot { display: inline-block; width: 8px; height: 8px; border-radius: 50%; margin-right: 6px;
flexshrink: 0; }
  .dot-c { background: #16a34a; }
  .dot-w { background: #dc2626; }
 </style>
</head>
<body>
<div class="quiz-card shadow-sm">

 <!-- Quiz Screen -->
 <div id="quiz-screen">
  <div class="progress">
   <div class="progress-bar" id="prog-fill" style="width:0%"></div>
  </div>
  <div class="q-meta">
   <span class="q-label" id="q-label">Question 1 of 10</span>
   <span class="score-pill" id="score-pill">Score: 0</span>
  </div>
  <p class="q-text" id="q-text"></p>
  <div id="options"></div>
  <div class="feedback-box" id="feedback"></div>
  <button class="btn btn-dark next-btn mt-1" id="next-btn" onclick="nextQ()">Next question
&rarr;</button>
 </div>

 <!-- Result Screen -->
 <div id="result-screen" style="display:none;"></div>

</div>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></scri  pt>
<script> const questions = [
 { q: "Which planet is closest to the Sun?",          opts:
["Venus","Mercury","Mars","Earth"],                ans: 1 },  { q:
"What is the chemical symbol for water?",         opts:
["H2O","CO2","NaCl","O2"],                        ans: 0 },
```

```
  { q: "How many continents are there on Earth?",          opts:
["5","6","7","8"],                          ans: 2 },
  { q: "Who painted the Mona Lisa?",                    opts: ["Michelangelo","Raphael","Leonardo da
Vinci","Caravaggio"], ans: 2 },
  { q: "What is the capital of Japan?",                  opts:
["Osaka","Kyoto","Hiroshima","Tokyo"],              ans: 3 },  { q: "How
many sides does a hexagon have?",            opts: ["5","6","7","8"],
ans: 1 },
  { q: "What gas do plants absorb from the atmosphere?",   opts: ["Oxygen","Nitrogen","Carbon
dioxide","Hydrogen"],     ans: 2 },
  { q: "Which is the largest ocean on Earth?",            opts:
["Atlantic","Indian","Arctic","Pacific"],           ans: 3 },
  { q: "What is 12 × 12?",                          opts: ["124","140","144","148"],                          ans: 2 },  { q:
"Which country invented the printing press?", opts:
["China","Italy","Germany","France"],               ans: 2 }
];

const KEYS = ["A","B","C","D"]; let cur = 0, score = 0,
answered = false, userAnswers = [];

function render() {   const q = questions[cur];   document.getElementById("qlabel").textContent
= `Question ${cur+1} of
${questions.length}`;    document.getElementById("score-pill").textContent =
`Score:     ${score}`;             document.getElementById("prog-fill").style.width       =
`${Math.round((cur/questions.length)*100)}%`;
document.getElementById("qtext").textContent                          =                    q.q;
document.getElementById("feedback").textContent                       =                    "";
document.getElementById("feedback").className            =           "feedback-box";
document.getElementById("next-btn").style.display = "none";   answered = false;

  const opts = document.getElementById("options");   opts.innerHTML =
"";
  q.opts.forEach((o, i) => {      const b =
document.createElement("button");      b.className
= "opt-btn";
    b.innerHTML = `<span class="key-badge">${KEYS[i]}</span>${o}`;
    b.onclick = () => select(i, b);
    opts.appendChild(b);
  });
}

function select(i, btn) {   if (answered) return;   answered = true;   const q = questions[cur];
document.querySelectorAll(".opt-btn").forEach(b =>
```

```
b.disabled = true);   const fb = document.getElementById("feedback");   if (i === q.ans) {
score++;     btn.classList.add("correct");       fb.innerHTML = `<svg width="16"
height="16" viewBox="0 0 20 20" fill="none"><circle
cx="10" cy="10" r="10" fill="#16a34a"/><path d="M6 10l3 3 5-5" stroke="#fff" strokewidth="2"
stroke-linecap="round" stroke-linejoin="round"/></svg> Correct!`;    fb.className = "feedbackbox
correct";   userAnswers.push({ correct: true, chosen: i }); } else {     btn.classList.add("wrong");
document.querySelectorAll(".optbtn")[q.ans].classList.add("correct");
   fb.innerHTML = `<svg width="16" height="16" viewBox="0 0 20 20" fill="none"><circle  cx="10"
cy="10" r="10" fill="#dc2626"/><path d="M7 7l6 6M13 7l-6 6" stroke="#fff" strokewidth="2" stroke-
linecap="round"/></svg> Incorrect — answer is
<strong>${KEYS[q.ans]}</strong>`;      fb.className
= "feedback-box wrong";
   userAnswers.push({ correct: false, chosen: i });
 }
 document.getElementById("score-pill").textContent = `Score: ${score}`;    const nb
= document.getElementById("next-btn");
nb.style.display = "inline-block";
 nb.textContent = cur < questions.length - 1 ? "Next question →" : "See results →";
}

function nextQ() {
 cur++;
 if (cur >= questions.length) showResult();
 else render();
}

function showResult() {   document.getElementById("quiz-
screen").style.display = "none";   const pct =
Math.round((score / questions.length) * 100);   let emoji = " 🎯 ", title = "Well done!", msg = "You have
a solid grasp of these topics.";   if (pct === 100) { emoji = " ⏳ "; title = "Perfect score!"; msg  = "You
nailed every question. Outstanding!"; }   else if (pct >= 80) { emoji = " 🌟 "; title = "Great job!"; msg =
"Almost perfect — impressive  knowledge!"; }   else if (pct < 50) { emoji = " 📚 "; title = "Keep
practising!"; msg = "Review the topics below and  try again."; }

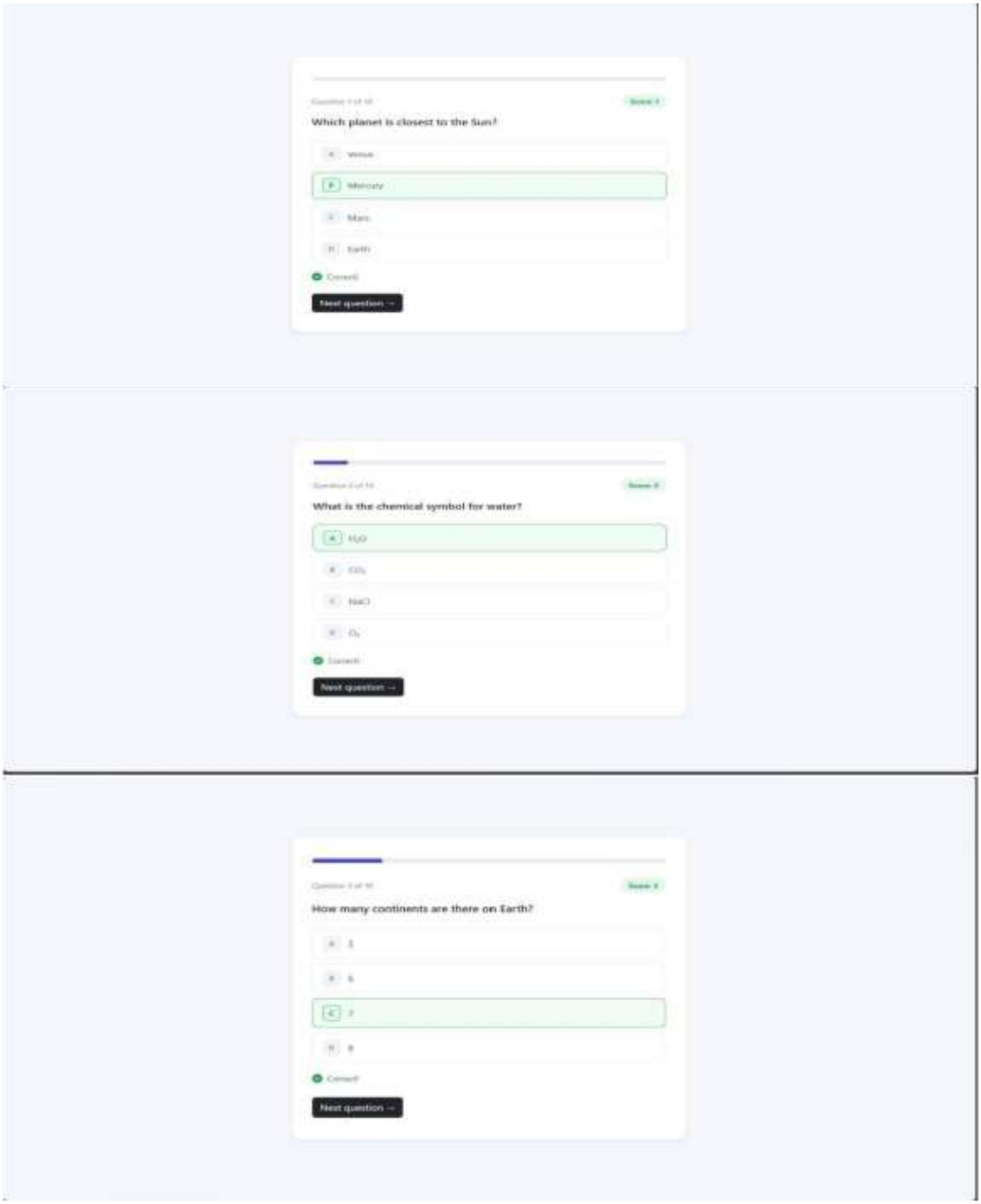
 let reviewHTML = `<div class="mt-4"><p class="text-muted" style="font-size:13px;font-
weight:600;margin-bottom:8px;">QUESTION REVIEW</p>`;   questions.forEach((q, i)
=> {     const ua = userAnswers[i];       reviewHTML += `<div class="review-
item d-flex align-items-start gap-2">
    <span class="dot mt-1 ${ua.correct ? 'dot-c' : 'dot-w'}"></span>
    <div>
      <div style="font-size:13px;font-weight:600;color:#1e293b;">${q.q}</div>
      <div style="font-size:13px;color:#64748b;">${ua.correct ? `Correct — ${q.opts[q.ans]}` :
`You chose: ${q.opts[ua.chosen]}  |  Answer: ${q.opts[q.ans]}`}</div>        </div></div>`;
 });
 reviewHTML += `</div>`;
```

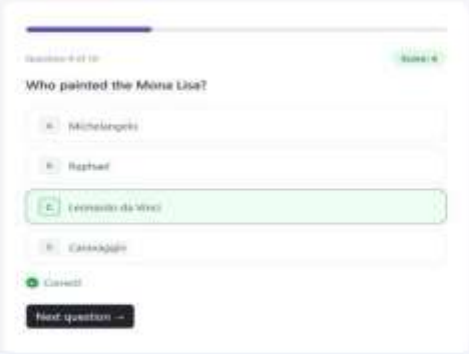
```
 const rs = document.getElementById("result-screen");
 rs.style.display = "block";   rs.innerHTML =
`
   <div class="text-center mb-3">
    <span class="result-emoji">${emoji}</span>
    <h4 class="fw-bold mb-1" style="color:#1e293b;">${title}</h4>
    <p class="text-muted" style="font-size:14px;">${msg}</p>
   </div>
   <div class="row g-2 mb-3">
    <div class="col-4"><div class="stat-box"><div
class="statval">${score}/${questions.length}</div><div  class="stat-lbl">Correct</div></div></div>
<div class="col-4"><div class="stat-box"><div class="stat-val">${pct}%</div><div
class="statlbl">Score</div></div></div>
    <div class="col-4"><div class="stat-box"><div class="stat-val">${questions.length -
score}</div><div class="stat-lbl">Missed</div></div></div>
   </div>
   <div class="text-center mb-3">
    <button class="btn btn-dark px-4" onclick="restart()">Try again</button>      </div>
${reviewHTML}`;
}

function restart() {
 cur = 0; score = 0; answered = false; userAnswers = []; document.getElementById("result-screen").style.display = "none";
document.getElementById("quiz-screen").style.display = "block";   render();
}

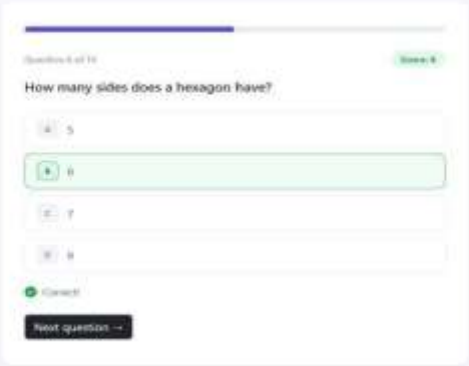
render(); </script>
</body>
</html>
```

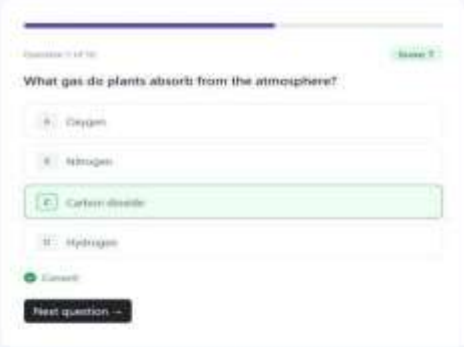
FINAL OUTPUT :

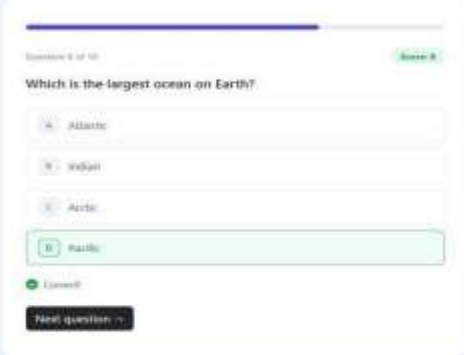



Who painted the Mona Lisa?
Michelangelo
Raphael
Leonardo da Vinci
Caravaggio
Correct!
Next question →

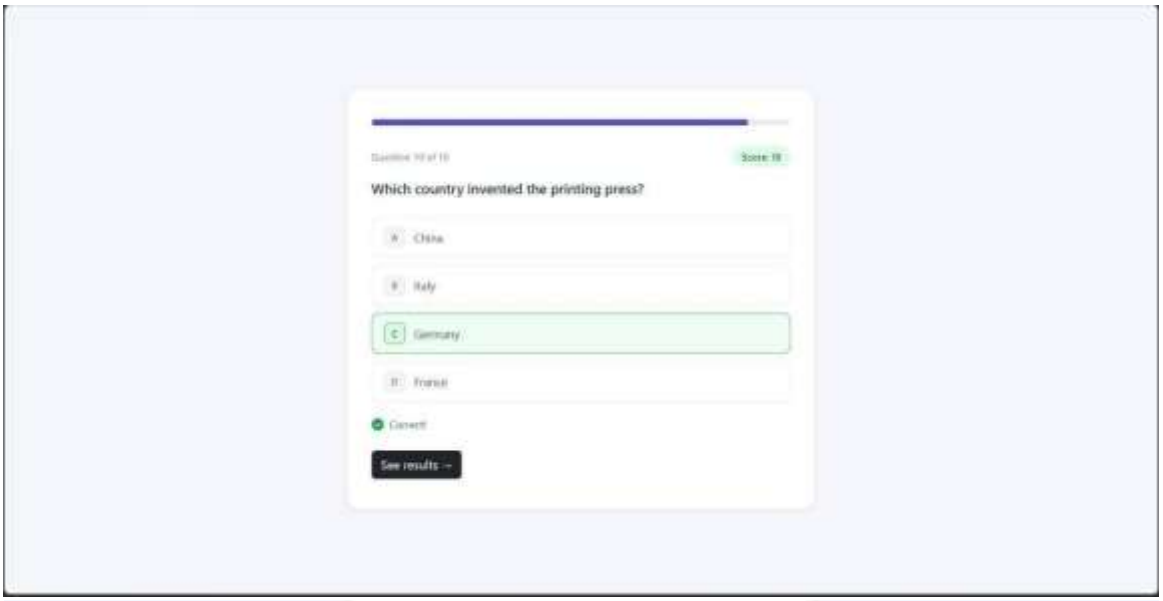
What is the capital of Japan?
Osaka
Kyoto
Hiroshima
Tokyo
Correct!
Next question →

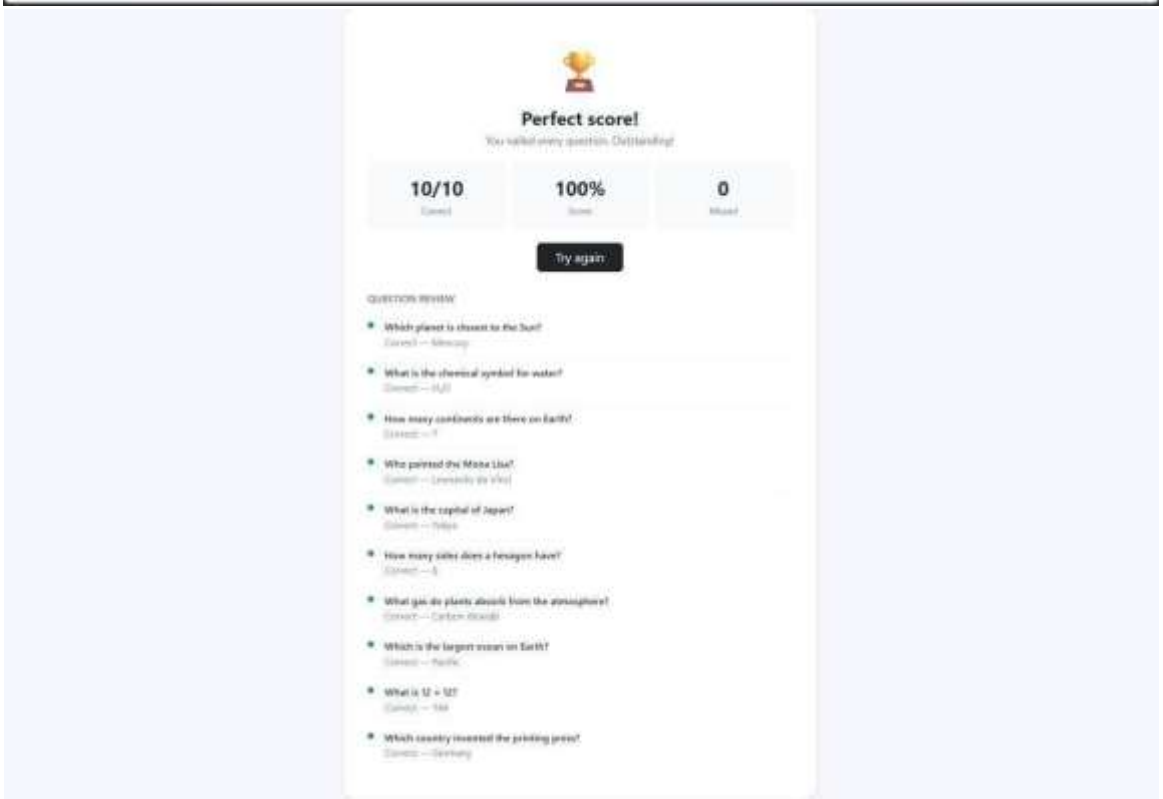

How many sides does a hexagon have?
5
6
7
8
Correct!
Next question →


What gas do plants absorb from the atmosphere?
Next question →


Which is the largest ocean on Earth?
Next question →

What is 12 × 12?
Next question →


Which country invented the printing press?
A China
B Italy
C Germany
D France
Correct!
See results →


Perfect score!
10/10
100%
0
Try again
QUESTION REVIEW